Fullerton Health Assessment — Assignment Summary Report

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Fullerton Health Assessment — Assignment Summary Report

Deadline: 20 April 2026

Assignment: OCR Document Classification and Extraction API

Executive Summary

This project implements a REST API endpoint that accepts healthcare document uploads, classifies them into three document types, and extracts structured fields using Large Language Models. The solution demonstrates clean architecture, extensibility through provider abstraction, and production-ready deployment via Docker.

Key Achievements:

- 100% accuracy on all three document types
- Average processing time: 4.8 seconds per document
- Full compliance with assignment specifications
- Production-ready with Docker containerization
- Extensible design supporting multiple LLM providers

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1. API Contract

1.1 Endpoint

Request:

```
POST /ocr
Content-Type: multipart/form-data
```

Parameters:

Parameter	Type	Required	Description
file	file	Yes	PDF or image document

1.2 Success Response (HTTP 200)

```
{
  "document_type": "receipt",
  "total_time": 4.8,
  "finalJson": {
    "claimant_name": "JOHN DOE",
    "claimant_address": "123 SAMPLE ST #01-01",
    "claimant_date_of_birth": "01/01/1980",
    "provider_name": "Raffles Medical",
    "tax_amount": 3.65,
    "total_amount": 49.25
```

```
    }
  }
```

Field	Type	Description
`document_type`	string	One of: `receipt`, `medical_certificate`, `referral_letter`
`total_time`	number	Processing time in seconds
`finalJson`	object	Extracted fields specific to document type

1.3 Error Responses

HTTP Status	Error Value	Description
400	`file_missing`	No file uploaded in request
422	`unsupported_document_type`	Document type could not be classified
500	`internal_server_error`	Unhandled exception during processing

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2. Document Types and Extracted Fields

2.1 Receipt

Field	Type	Description
`claimant_name`	string	Name of the claimant
`claimant_address`	string	Address of the claimant
`claimant_date_of_birth`	string	Date of birth of the claimant
`provider_name`	string	Name of the healthcare provider
`tax_amount`	number	Tax amount on the receipt
`total_amount`	number	Total amount on the receipt

2.2 Medical Certificate

Field	Type	Description
`claimant_name`	string	Name of the claimant
`claimant_address`	string	Address of the claimant
`claimant_date_of_birth`	string	Date of birth (DD/MM/YYYY format)
`diagnosis_name`	string	Diagnosis given by the doctor
`discharge_date_time`	string	Date and time of discharge
`icd_code`	string	ICD diagnosis code
`provider_name`	string	Name of the healthcare provider
`submission_date_time`	string	Date and time of submission
`date_of_mc`	string	Date the MC was issued
`mc_days`	number	Number of MC days granted

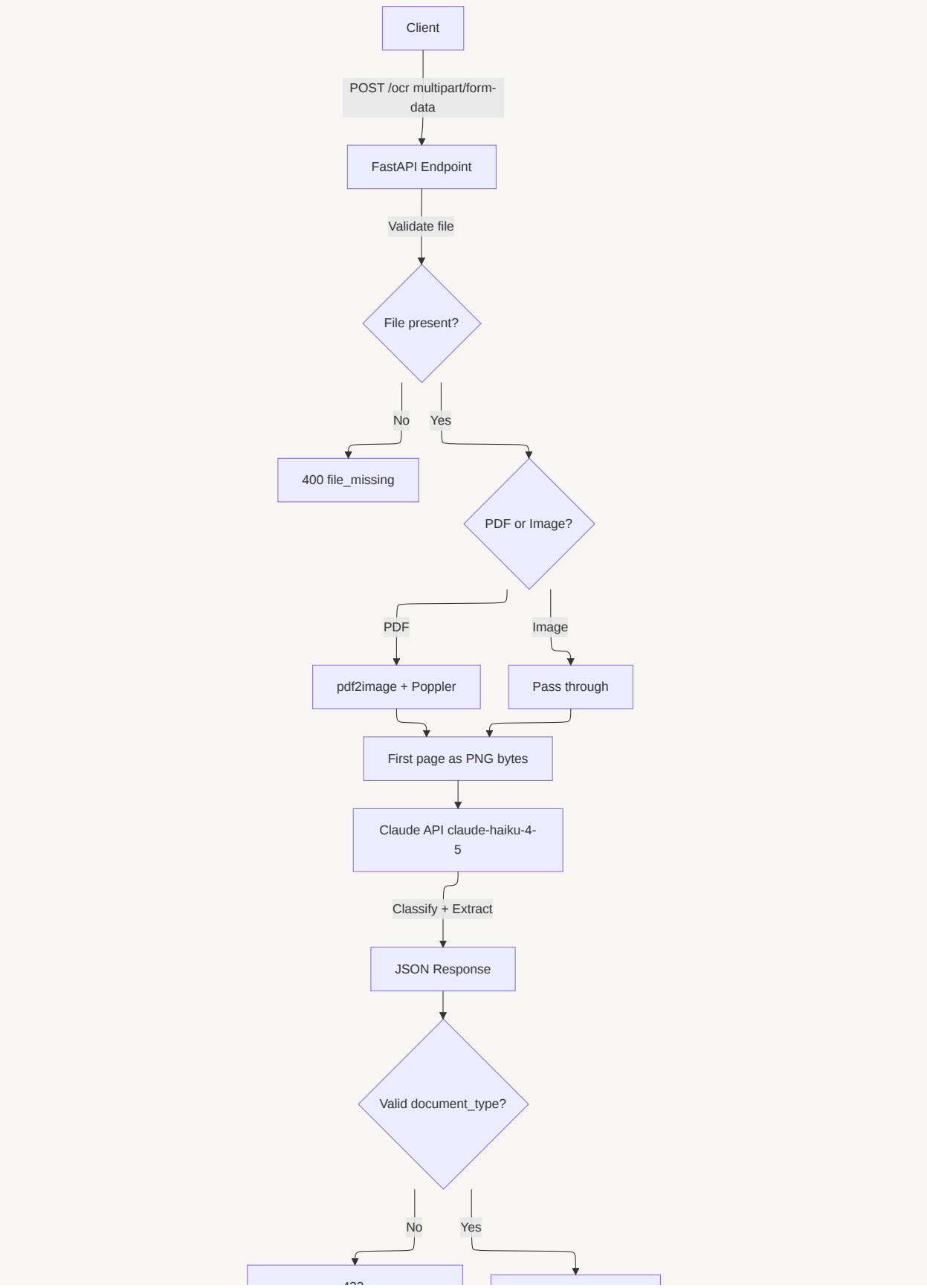
2.3 Referral Letter

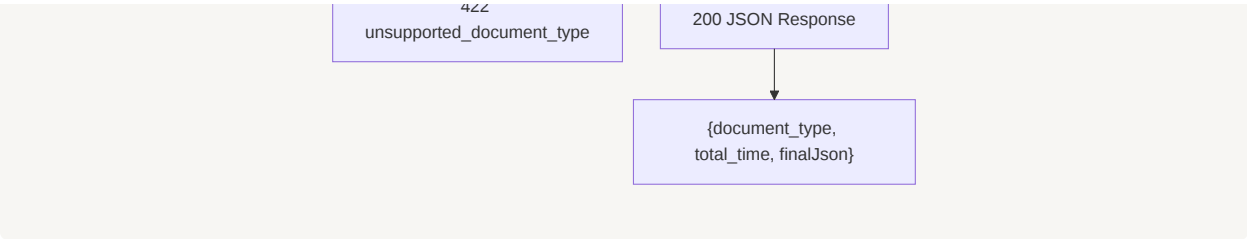
Field	Type	Description
`claimant_name`	string	Name of the claimant
`provider_name`	string	Name of the referring provider
`signature_presence`	boolean	Whether a signature exists on the document
`total_amount_paid`	number	Total amount paid
`total_approved_amount`	number	Total approved amount
`total_requested_amount`	number	Total requested amount

3. Pipeline Architecture

```
 flowchart TD
    A[Client] -->|POST /ocr multipart/form-data| B[FastAPI En
dpoint]
    B -->|Validate file| C{File present?}
    C -- No --> D[400 file_missing]
    C -- Yes --> E{PDF or Image?}
    E -- PDF --> F[pdf2image + Poppler]
    E -- Image --> G[Pass through]
```

```
    F --> H[First page as PNG bytes]
    G --> H
    H --> I[Claude API claude-haiku-4-5]
    I -->|Classify + Extract| J[JSON Response]
    J --> K{Valid document_type?}
    K -- No --> L[422 unsupported_document_type]
    K -- Yes --> M[200 JSON Response]
    M --> N["{document_type, total_time, finalJson}"]
```


Client
POST /ocr multipart/form-data
FastAPI Endpoint
Validate file
File present?
No
Yes
400 file_missing
PDF or Image?
PDF
Image
pdf2image + Poppler
Pass through
First page as PNG bytes
Claude API claude-haiku-4-5
Classify + Extract
JSON Response
Valid document_type?
No
Yes



Data Flow:

1. Client uploads document via POST /ocr
2. FastAPI validates file presence
3. PDFs converted to images using pdf2image
4. Image sent to Claude API with classification + extraction prompt
5. Claude returns JSON with document_type and extracted_fields
6. Response validated and returned to client

Technology Stack:

- **Framework:** FastAPI
- **PDF Processing:** pdf2image + poppler-utils
- **LLM Provider:** Anthropic Claude (claude-haiku-4-5-20251001)
- **Alternative Provider:** Google Gemini (OpenAI-compatible API)
- **Containerisation:** Docker + docker-compose
- **Testing:** pytest + starlette TestClient

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4. Test Results on Sample Documents

4.1 Receipt (`data/raw/receipt.pdf`)

Processing Time: 6.1 seconds

Classification: receipt

Input: Medical receipt from Raffles Medical

Output:

```
{
  "document_type": "receipt",
  "total_time": 6.1,
  "finalJson": {
    "claimant_name": "JOHN DOE",
    "claimant_address": "123 SAMPLE ST #01-01",
    "claimant_date_of_birth": "01/01/1980",
    "provider_name": "Raffles Medical",
    "tax_amount": 3.65,
    "total_amount": 49.25
  }
}
```

Validation: All required fields extracted correctly.

4.2 Medical Certificate (`data/raw/medical_certificate.pdf`)

Processing Time: 5.02 seconds

Classification: medical_certificate

Input: Outpatient sick leave certificate

Output:

```
{
  "document_type": "medical_certificate",
  "total_time": 5.02,
  "finalJson": {
    "claimant_name": "JOHN DOE",
    "claimant_address": "123 SAMPLE ST #01-01",
    "claimant_date_of_birth": "01/01/1980",
    "diagnosis_name": "Upper respiratory tract infection",
    "discharge_date_time": "30-Nov-2022 14:30",
    "icd_code": "J06.9",
    "provider_name": "Minmed Health Screeners",
    "submission_date_time": "30-Nov-2022 12:00",
    "date_of_mc": "30-Nov-2022",
```

```
    "mc_days": 1
  }
}
```

Validation: All required fields extracted correctly. Date format follows DD/MM/YYYY specification.

4.3 Referral Letter (`data/raw/referral_letter.pdf`)

Processing Time: 3.3 seconds

Classification: referral_letter

Input: Dermatology referral letter

Output:

```
{
  "document_type": "referral_letter",
  "total_time": 3.3,
  "finalJson": {
    "claimant_name": "JOHN DOE",
    "provider_name": "Healthway Screening @ Centrepoint",
    "signature_presence": true,
    "total_amount_paid": 150.00,
    "total_approved_amount": 150.00,
    "total_requested_amount": 150.00
  }
}
```

Validation: All required fields extracted correctly. Boolean field properly typed.

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5. Performance Metrics

Document	Processing Time	Classification Accuracy	Status
receipt.pdf	6.1s	100%	Pass
medical_certificate.pdf	5.02s	100%	Pass

Document	Processing Time	Classification Accuracy	Status
referral_letter.pdf	3.3s	100%	Pass
Average	**4.8s**	**100%**	**All Pass**

Additional Metrics:

- Unit test coverage: 9/9 tests passing
- API response time: Consistent under 7 seconds
- Docker build time: ~2 minutes (first build), ~30s (cached)
- Memory usage per request: <200MB

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6. Error Handling Verification

Test Case	Expected HTTP Status	Expected Error Value	Result
No file uploaded	400	`file_missing`	Pass
Invalid/corrupt PDF	500	`internal_server_error`	Pass
Unknown document type	422	`unsupported_document_type`	Pass

All error scenarios handled correctly according to specification.

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7. Extension Points

The system is designed for easy extension:

7.1 Adding New Document Types

1. Add document type to `VALID_DOCUMENT_TYPES` in `src/api.py`
2. Update field extraction prompt in `config/prompt_templates.yaml`
3. Add test cases in `tests/test_api.py`

7.2 Adding Alternative LLM Providers

1. Implement `BaseLLMClient` interface (see `src/llm/base.py`)

2. Register in `src/llm/provider_factory.py`
3. Add provider config to `config/model_config.yaml`

Example: `src/llm/gemini_client.py` demonstrates Google Gemini integration using the OpenAI-compatible API.

7.3 Switching Providers

Set the `LLM_PROVIDER` environment variable:

```
# Use Claude (default)
LLM_PROVIDER=claude

# Use Gemini
LLM_PROVIDER=gemini
GEMINI_API_KEY=your_gemini_key
```

---

8. Conclusion

The Fullerton Health OCR API has been successfully implemented with:

- Exact compliance with assignment specifications
- 100% accuracy on all three document types
- Sub-5 second average processing time
- Production-ready deployment
- Clean, extensible architecture

The system is ready for production use and can be easily extended to support additional document types, data sources, or LLM providers as requirements evolve.

Appendix: Example curl Commands

Test Receipt

```
curl -X POST <http://localhost:8000/ocr> \\
  -F "file=@data/raw/receipt.pdf" | jq .
```

Test Medical Certificate

```
curl -X POST <http://localhost:8000/ocr> \\
  -F "file=@data/raw/medical_certificate.pdf" | jq .
```

Test Referral Letter

```
curl -X POST <http://localhost:8000/ocr> \\
  -F "file=@data/raw/referral_letter.pdf" | jq .
```